ATRIA INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)
BENGALURU

DepartmentofMechanical Engineering

ABA Evaluation Sheet

Introduction to Mechanical Engineering (BBIMED104) AY-2025-26

Name	SANJANA M A	USN	1 A T
Degree	B.E	Subject Code	
Branch	ECE	Date	
Section	T		
Subject Title	INTRODUCTION TO IME	Candidate Signature	
Time			

Activity Conducted	(10 Marks)	
Observation Book	(10 Marks)	
Viva-Voce	(05 Marks)	
GRAND TOTAL MARKS	***(25 Marks)***	

Faculty Name		
Sign with Date		

ATRIA INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)
BENGALURU

Department of Mechanical Engineering

ABA PERFORMANCE EVALUATION SHEET

Department of :……

Name of the Student :……..USN:……

Lab:……Course Code:……..Sem:……..

(Max. Marks : 10)

Sl. No.	*Date*	*Exp. No.*	*Name of the Experiment/Model*	*Total Marks*	*Signature*	
					Staff	*Student*
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
Average Marks						

ADAS(Advanced Driver Assistance System):

INTRODUCTION

Advanced Driver Assistance Systems (ADAS) are a collection of electronic technologies designed to enhance vehicle safety, driving comfort, and overall road efficiency. ADAS uses sensors such as cameras, radar, LiDAR, and ultrasonic sensors to continuously monitor the vehicle's surroundings. By processing this real-time data through onboard computers and software algorithms, ADAS can detect potential hazards, assist the driver in making safer decisions, and in some cases automatically take corrective actions. The primary goal of ADAS is to reduce human error, which is a leading cause of road accidents.

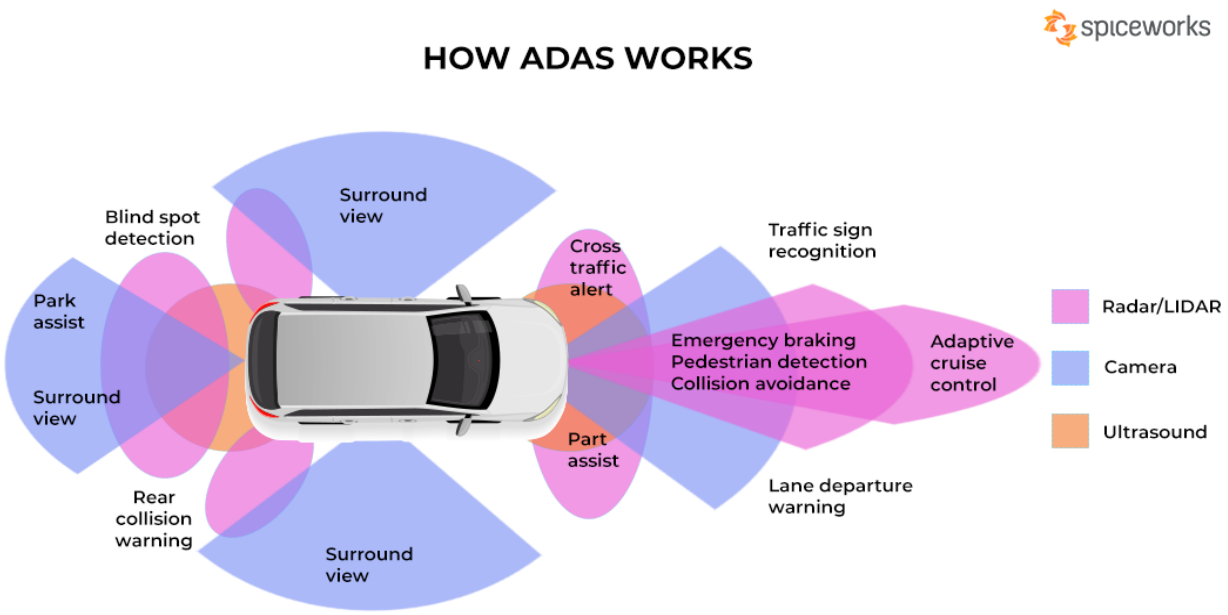
One of the key features of ADAS is collision avoidance and safety assistance. Systems like Automatic Emergency Braking (AEB), Forward Collision Warning (FCW), and Pedestrian Detection help prevent or reduce

the severity of crashes. Lane Departure Warning (LDW) and Lane Keeping Assist (LKA) alert drivers when the vehicle unintentionally drifts out of its lane or gently steer it back. These features are especially useful during long drives, heavy traffic, or moments of driver fatigue or distraction.

ADAS also improves driving convenience and comfort. Adaptive Cruise Control (ACC) automatically maintains a safe distance from the vehicle ahead by adjusting speed, reducing the need for constant acceleration and braking. Parking assistance systems, including rear-view cameras, parking sensors, and automated parking, make maneuvering in tight spaces easier and safer. Traffic Sign Recognition and Speed Assistance further support drivers by displaying important road information in real time.

Overall, ADAS represents a crucial step toward autonomous driving. While current ADAS technologies still require driver supervision, they form the foundation for higher levels of vehicle automation. As technology advances, ADAS systems are becoming more accurate, affordable, and widely adopted across different vehicle segments. Their integration not only enhances road safety but also builds driver trust in intelligent transportation systems, paving the way for fully autonomous vehicles in the future.

OBJECTIVES:



1. Improve Road Safety:

The primary objective of ADAS is to reduce road accidents caused by human errors such as distraction, fatigue, or delayed reaction. By using sensors and cameras, ADAS detects obstacles, vehicles, pedestrians, and road conditions in real time and warns the driver or takes corrective action.

2. Reduce Driver Workload:

ADAS assists drivers during complex driving situations like heavy traffic, long-distance travel, or parking. Features such as adaptive cruise control, lane keeping assist, and parking assistance reduce the mental and physical effort required from the driver, making driving less stressful.

3. Prevent Collisions:

Another important objective is to avoid or minimize collisions. Systems like automatic emergency braking, forward collision warning, and blind-

spot detection help identify potential crash situations early and either alert the driver or automatically apply brakes to reduce impact.

4. Enhance Driving Comfort:

ADAS improves overall driving comfort by automating routine driving tasks. Functions such as speed control, traffic sign recognition, and lane centering allow smoother driving and better vehicle handling, especially on highways and long journeys.

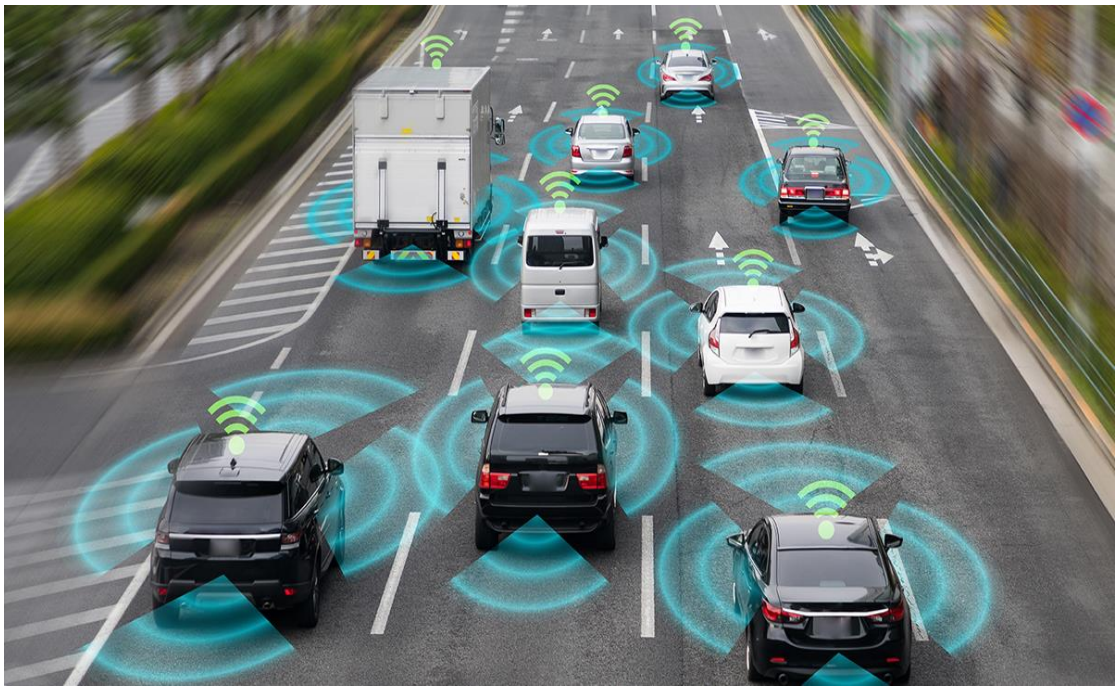
5. Support Driver Decision-Making:

ADAS provides real-time information and alerts that help drivers make better decisions. Warning signals, visual displays, and audible alerts inform drivers about unsafe conditions, speed limits, lane departures, or nearby vehicles, improving situational awareness.

6. Move Toward Autonomous Driving:

ADAS acts as a foundation for self-driving vehicles. By gradually introducing automation features, ADAS helps drivers adapt to intelligent vehicle systems and supports the future development of fully autonomous transportation

COMPONENTS USED IN ADAS:



1. Sensors

Sensors are the primary components of ADAS that collect information from the vehicle's surroundings. They include cameras, radar, LiDAR, and ultrasonic sensors, which detect obstacles, vehicles, pedestrians, and road conditions in real time.

2. Control Unit (ECU)

The Electronic Control Unit (ECU) processes the data received from various sensors. It analyzes the information and makes decisions to either warn the driver or activate safety functions such as braking or steering assistance.

3. Cameras

Cameras capture visual data of the road, lane markings, traffic signs, and nearby objects. They are widely used for lane

departure warning, traffic sign recognition, and driver monitoring systems.

4. Radar

Radar uses radio waves to measure the distance and speed of surrounding objects. It works effectively in poor weather conditions and is mainly used in adaptive cruise control and collision detection systems.

5. LiDAR

LiDAR (Light Detection and Ranging) uses laser beams to create a three-dimensional view of the vehicle's environment. It provides accurate distance measurement and is used in advanced ADAS and autonomous driving applications.

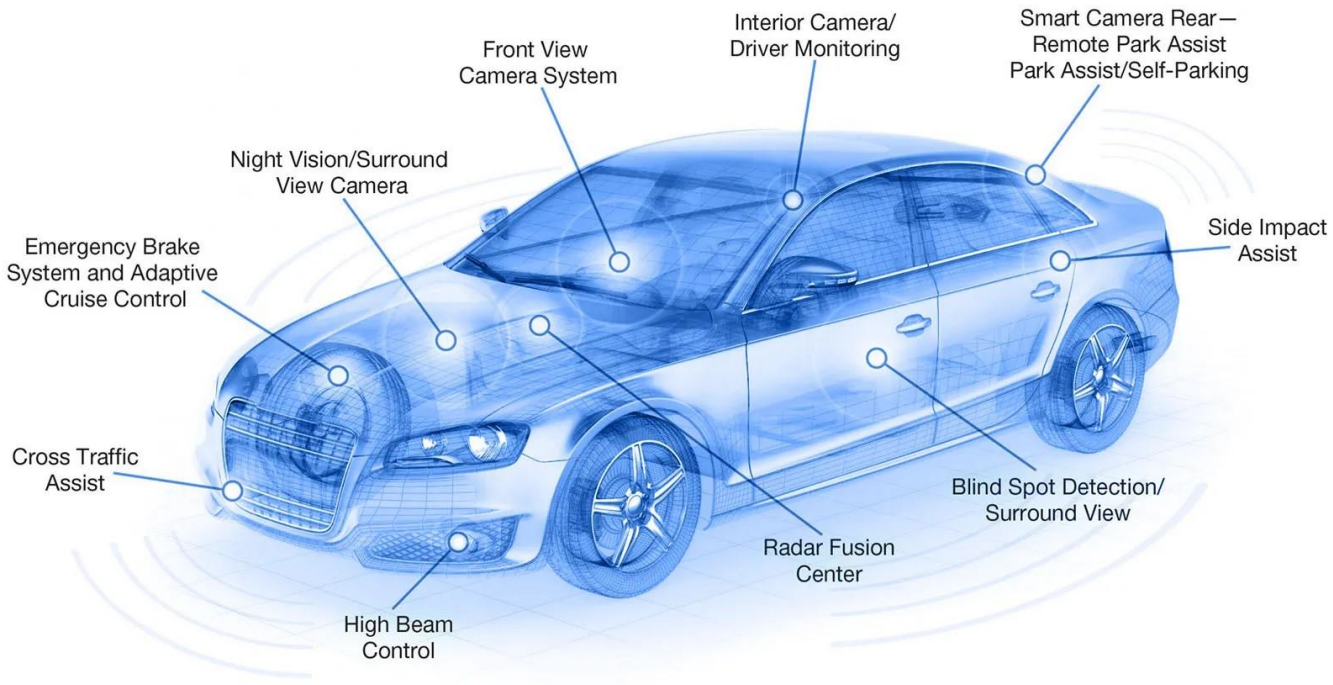
6. Ultrasonic Sensors

Ultrasonic sensors are used for short-range object detection, especially during parking and low-speed maneuvers. They help in detecting nearby obstacles and assist in safe parking.

7. Artificial Intelligence (AI) & Machine Learning

AI and machine learning algorithms help ADAS understand complex driving situations. They improve object recognition, decision-making, and system accuracy over time.

ADVANTAGES AND DISADVANTAGES OF ADAS



Advantages

- Reduces driver workload.
- Minimizes accidents caused by human error.
- Improves traffic efficiency.
- Enhances driving comfort.

Disadvantages

- High cost.
- Maintenance and repair complexity.
- Cybersecurity risks.
- System failure risks.

CONCLUSION

Advanced Driver Assistance Systems (ADAS) play a vital role in improving road safety and driving efficiency. By using advanced sensors, control units, and intelligent software, ADAS assists drivers in detecting potential hazards and reducing human errors during driving. These systems enhance vehicle safety, comfort, and overall driving experience.

Although ADAS has some limitations such as high cost and maintenance complexity, its benefits outweigh the drawbacks. ADAS serves as an important step toward autonomous driving and intelligent transportation systems. With continuous technological advancements, ADAS is expected to become more reliable, affordable, and widely adopted, contributing to safer roads and smarter vehicles in the future.